

FOUNDATION MODULE

Programming Foundations

A Comprehensive Overview of JavaScript & Python

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Full Stack Web Development | SMIT

SECTION 01

JavaScript Interaction

Learning how to communicate with the user through Alerts, Console logs, and Input Prompts.

The `alert()` Method

Immediate Notification

The `alert()` function displays a dialog box with a message to the user.

It is a "blocking" function, meaning the user must interact with the pop-up before the rest of the script continues to run.

```
alert("This is an Important Message!");
```


The `console.log()` Method

Internal Debugging

`console.log()` outputs data silently to the browser console without interrupting the user experience.

It is primarily used by developers to print results, check variable values, and find errors during the development process.

```
console.log("Debugging System ... OK");
```


The `prompt()` Method

Capturing User Input

The `prompt()` function opens a pop-up box that allows the user to type in their own information.

Important: No matter what the user types (even if it's a number), JavaScript always receives the input as a **String**.

```
let name = prompt("What is your name?");
```


Python: The `print()` Function

Terminal Output

In Python, we display information directly to the terminal or console using the simple `print()` function.

Python's syntax is designed for high readability, making output generation straightforward and intuitive.

```
print("Outputting results to terminal... ")
```


Variables: Working with Strings

A

Text Storage

Strings are variables that store plain text data like names or addresses.

”

Quotation Rule

Text must hamesha be wrapped in `" "` or `' '` to be treated as a string.

</>

Example

```
let city =  
"Karachi";
```


Variables: Working with Numbers

Numerical Values

Numbers represent whole integers or decimals used for math.

Rule: Do not use quotes! Writing `"25"` tells the computer it is text, and you cannot perform math on it.

```
let score = 100; // Ready for calculations
```


Variable Naming Conventions

CONSTRAINT	LEGAL NAMES	ILLEGAL NAMES
Start Char	<code>user</code> , <code>_id</code>	<code>1user</code> (Start with digit)
Whitespace	<code>fullName</code>	<code>full name</code> (No spaces)
Special Chars	<code>total_sum</code>	<code>my@var</code> (No symbols)

Math Operators

+

Addition

$5 + 5 = 10$

*

Multiplication

$4 * 3 = 12$

%

Modulus (%)

Returns the **remainder**
($10 \% 3 = 1$).

Eliminating Ambiguity

BODMAS / PEMDAS

To ensure calculations run in the correct order, we use **Brackets `()`**. This removes any confusion for the computer.

$$5 + 2 * 3 = 11$$

$$(5 + 2) * 3 = 21$$

Concatenating Text Strings

Joining Strings

The `+` operator can also be used to join two or more text strings together into one sentence.

```
let fullName = "Maheen" + " " + "Abbasi";
```

Result: "Maheen Abbasi"

Decision Making Logic

Conditionals

Using `if` and `else`, the program decides which code block to execute based on a condition.

```
if (marks ≥ 50) {  
    alert("Pass");  
} else {  
    alert("Fail");  
}
```


Thank You

Questions?

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SMIT Foundations Module Presentation